

DISCREPANCY BETWEEN PREPARATION AND PRACTICE IN TEACHER EDUCATION: AN ECUADORIAN STUDY

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Abstract— A university in Ecuador requested assistance with curriculum revision so as to make its teacher training program more effective. In order to focus on real rather than perceived problems, a study of the curriculum was carried out. Over 200 graduates of the program were surveyed using a written questionnaire based on the content of the Faculty of Education curriculum. Results of the survey indicated a significant gap between preparation and practice. The major problems with preparation were identified as content emphasis and implementation rather than curriculum structure as perceived originally by university authorities.

Faculties of education, as professional schools, have as one of their functions the task of training practitioners to perform effectively in classrooms. This means imparting competencies to their students to enable them to cope with the wide variety of conditions in which teachers must work, including operating within a specific social, economic, and political context.

When a faculty, or any other institution for that matter, recognizes that it is not fulfilling its role as effectively as it might, it should consider undertaking some form of evaluation in order to assess, in a systematic way, its areas of strengths and weaknesses prior to initiating changes. Otherwise there is danger that solutions may be applied to problems that do not exist. Donald (in press) clearly defines evaluation as,

the process of determining value or merit. It consists of methods for the systematic collection and analysis of information with the aim of improving the quality of human services. It is an applied social science which examines the effectiveness and efficiency of these services and enables adaptations to be made to them. The essential features of evaluation as a form of inquiry are its attention to performance or achievement, its use in decision-making, and its concern with both ethics and utility.

However, evaluation in its various forms can be a complex undertaking. The literature on this topic, Provus (1971), Stufflebeam et al. (1971), Stake (1975), and Donald (in press), shows that there are specific processes to meet a variety of needs and conditions, and that most of these require considerable time, and in many cases the use of relatively involved procedures. The total process often requires a certain degree of sophistication on the part of those seeking the evaluation since they must be active participants in the undertaking. In addition, evaluation and its various procedures must take into account the institution itself in its social, economic, and cultural setting, and its ability to cope with the demands of the process as well as to implement the recommendations derived from its conclusions. This is not easy under any circumstances, and it becomes far more difficult when the process is attempted in an institution in a developing country where the personnel involved may be totally unfamiliar with systematic evaluation and its specific procedures.

The authors found themselves in this difficult position when they were requested by a university in Ecuador to provide professional develop-

ment activities for the professors of its Faculty of Education as well as to assist, within somewhat vague parameters, with curriculum reform in that faculty. Given an ill-defined mandate, local conditions, limited time, and restricted financial and human resources, a limited program evaluation was conducted based on a variation of procedures that had been used in other circumstances. The aim was to provide the Faculty of Education with very basic and concrete data about its teacher training program, in order to recommend realistic and realizable actions that could be implemented to improve the quality of the training it was offering currently. This article is a report of the results of this limited program evaluation which revealed that the perceived problem with teacher preparation was not the actual problem.

The Problem

The institution concerned is a large (60,000 students) state university with a Faculty of Education of some 10,000 students. The Faculty was vaguely aware that its preparation of secondary school teachers was weak. Perceptions of the imperfections of training were based neither on hard data nor on systematic observation, but rather on opinions and intuition. Nevertheless, attention was focused on the program of studies as the crux of the perceived inadequacies; the administration of the Faculty of Education seemed determined to make major structural changes in its curriculum. They were uncertain however, as to how to make effective changes that could in fact be translated into the preparation of more competent practitioners.

Improving the training of practitioners is a process that may go beyond making structural changes in a program of studies. Proposing structural changes, however, may seem attractive on the surface. Such changes tend to be nonthreatening both to the administration and to the professorial staff, and the considerable activity involved, often cosmetic and not touching the real issues, can often be mistaken for progress.

From the outside, it appeared that the program of studies, by North American standards, was excessively inflexible and unimaginative and resembled what had been done in the indus-

trialized countries more than a generation ago. Nevertheless, this did not imply that an effective application of the program, or variations of it, could not produce teachers sufficiently competent to cope with the educational tasks they had to perform given their particular context. It was observed also that the administrative policies and practices of the university and of the Faculty of Education seemed rigid and were not always applied effectively. Furthermore, because of the financial situation both of the university and of the country, there was a serious lack of educational resources and working conditions were generally poor. Finally, the faculty itself was in need of professional development so as to keep abreast of recent educational theories and practices.

The task of assisting the Faculty of Education to make meaningful improvements in the way it was preparing secondary school teachers did not necessarily imply a total reform or reorganization of the system. In fact, given the current educational conditions in the country, developing a curriculum based on models from industrialized countries might do little toward making secondary school teachers function more effectively in their milieu. It seemed more realistic to identify and to focus on some critical issues that could be addressed within the context of available or obtainable resources. Making a limited number of specific, pertinent, and practical changes might have a sufficient impact to improve the training currently offered and make noticeable improvement in the actual performance of practitioners. Concrete evidence was gathered not only concerning the program of studies but also on the effectiveness of its application. This was done in a way that would be understood by and convincing to the Faculty of Education authorities, otherwise proposed recommendations would prove meaningless.

Development of a Plan of Action

The authors worked with a small team from the Faculty of Education which had been formed to proceed with a reform of the curriculum. However, as was stated above, there was no evidence to indicate that the perceived inadequacies in the preparation of teachers were solely or necessarily a curriculum prob-

lem. The team decided to investigate the problem by systematically surveying graduates of the program — practicing teachers trained by the Faculty of Education — in order to gather information about the curriculum and its application.

The decision to use former graduates of the program for the survey was based on two factors, evidence in the literature on evaluation, and on the authors' own successful experience in using former graduates and practitioners as a source of information about programs of study. The literature on evaluation supports the use of students or graduates, the consumers of programs, as a source of data. Provus (1971), for example, notes that the use of the opinion of those affected by a program has "the advantage of providing first hand accounts of program strengths and weaknesses from the perspective of those close to program operations" (p. 10). Donald (in press) agrees with this process and supports her argument by quoting another authority on evaluation: "Stake thus brought the focus of evaluation back to the consumers or concerned constituents of the program". Finally, Pratt (1980) writes: "Recent graduates are a fruitful and frequently overlooked source for commentary on the curriculum. They are now finding out how their learnings apply to life. Graduates were a major source of data in the research leading to the Swedish curriculum reforms of the 1960s" (p. 83).

Applying prescription to practice, the authors had based two earlier studies on the perceptions of former graduates and practitioners. Weston (1982) surveyed practicing instructional developers to analyze their competencies in relation to their training programs. Boulianne and Weston (1983) used a modification of this approach to analyze the relationship between a specific course content and the practices of classroom teachers. In both cases useful data were obtained regarding the relationship among competencies, course content, and the needs of practicing professionals.

Method

Subjects

The respondents were secondary school

teachers, graduates of the Faculty of Education, employed at 14 public and private schools which were representative of the various socioeconomic conditions in the metropolitan area.

Survey Instrument

Items for the questionnaire were derived from the professional component of the Faculty of Education curriculum. The curriculum, covered over a period of five years, is divided into three distinct sections: academic concentration, general culture, and pedagogical training. This latter section comprises 14 compulsory courses offered in a fixed sequence over the five-year period. It was this element of the program, the actual professional training component, which provided the items for the questionnaire (see Table 1).

Table 1

Professional Courses in the Faculty of Education

Year	Courses
I	General Pedagogy General Sociology
II	General Psychology General Didactics
III	Pedagogical Problems in Secondary Education Special Didactics Developmental and Adolescent Psychology Audio Visual Education
IV	Evaluation and Statistics School Legislation Psychopedagogy Orientation to Student Teaching
V	Methods and Techniques of Research History of Education

An initial list of professional competencies allegedly taught by the Faculty of Education was prepared from the objectives and content of the 14 education courses of the pedagogical section of the curriculum. The final list of competencies was reviewed, revised, and approved by the professors who taught each of these 14 courses in order to ensure that the competency statements reflected their content and context accurately. The final list was made up of 46 items divided into three categories: Instructional Process, items 1–25; Theories and Foundations,

items 26–36; Other Professional Attitudes and Activities, items 37–46 (see Table 2). These three broad categories were adapted from a taxonomy of educational competencies developed by Robinson et al. (1985). His 12 categories were finally collapsed to three to

avoid relatively complex organizations of competencies as well as terminology which might be unfamiliar to the target population.

Each competency was to be evaluated in two ways. First the respondents were to indicate, on a scale of 1–5 (1 = no importance; 5 = much im-

Table 2

List of Items Included in the Questionnaire (Abbreviated and Translated)

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|------|--|
| I. | Instructional Process |
| | 1. Prepare annual plan based upon official curriculum. |
| | 2. Annually update program of studies. |
| | 3. Select content and techniques based upon student factors. |
| | 4. Use the media as a source of instructional materials. |
| | 5. Plan different types of classes. |
| | 6. Prepare aligned course plans. |
| | 7. Formulate learning objectives. |
| | 8. Organize course content considering the time constraints. |
| | 9. Prepare evaluation of learning for objectives and content. |
| | 10. Plan field trips and other outside activities. |
| | 11. Create a favorable teaching–learning environment. |
| | 12. Motivate learning using appropriate techniques. |
| | 13. Motivate students to actively participate in learning. |
| | 14. Use group dynamics to make students more productive. |
| | 15. Control behavior based upon psychological principles. |
| | 16. Apply principles of psychology, logic and communication to teaching. |
| | 17. Establish mechanisms to maintain teacher–student communication. |
| | 18. Identify and procure help for students with problems. |
| | 19. Select and/or create instructional materials. |
| | 20. Use appropriately a-v materials and other resources. |
| | 21. Use teaching methods appropriate to objectives. |
| | 22. Carry out field trips and other outside activities. |
| | 23. Evaluate learning in accordance with objectives. |
| | 24. Analyze evaluation results. |
| | 25. Evaluate instruction. |
| | |
| II. | Foundations |
| | 26. Know the historical development of education. |
| | 27. Know the goals and principles of the educational system. |
| | 28. Know the difference between modern and traditional education. |
| | 29. Know the principle problems of pedagogy and the effect on education. |
| | 30. Know the role of the teacher, student and community in education. |
| | 31. Know the social context of the student. |
| | 32. Know the application of psychology to education. |
| | 33. Try to adapt teaching to student differences. |
| | 34. Know the fundamentals of instructional planning. |
| | 35. Know your rights and responsibilities as an educator. |
| | 36. Be familiar with laws relating to education. |
| | |
| III. | Other professional attitudes and activities |
| | 37. Analyze and search for solutions to educational problems. |
| | 38. Carry out educational research. |
| | 39. Evaluate your work in relation to national educational goals. |
| | 40. Promote improved use of a-v and traditional materials. |
| | 41. Promote educational innovation. |
| | 42. Be sensitive to adolescent motivation and conduct. |
| | 43. Demonstrate democratic attitudes in teaching. |
| | 44. Motivate students to be active members of society. |
| | 45. Transmit values important to the development of the country. |
| | 46. Demonstrate a serious attitude towards the teaching profession. |
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portance), the importance of the item in their actual practice as professionals. Second, on a similar 1–5 scale (1 = no preparation; 5 = much preparation), they were to rate the preparation they had received in the university to perform each of these skills.

The final questionnaire comprised three sections: demographic data, the 46 competencies, and five open-ended questions to supplement the list of competencies and to attempt to draw out other factors that might have had an impact on the training process. The five questions covered the following topics: the most outstanding aspects of studies in the Faculty of Education; the weakest elements of the program; lacunae in the program; factors which impeded a better preparation; omissions in the questionnaire.

The questionnaire was validated by experts from the Faculty of Education and was field tested with a group of 15 teachers.

Distribution and Retrieval of Questionnaire

There were 250 questionnaires distributed and administered by a team of 12 professors from the Faculty of Education trained to perform this task. In all, 207 questionnaires were collected, and of these, 204 were usable. The high response rate was in part attributable to the work of the team who carried out their task assiduously.

Results

Descriptive Data

The respondents represented a good cross-section of secondary school teachers of the region. The majority were certified teachers with the degree of Licenciado (a specialized undergraduate degree), had been teaching between 10 and 20 years, and represented the various academic specializations taught in the secondary schools. The great majority, 70%, were teaching in the public school system.

The means, on the scale from 1 to 5, for each of the 46 items are represented graphically both for *Importance* and *Preparation* (see Figure 1). The Grand Mean (M_g), 4.5 for *Importance* and 2.9 for *Preparation*, is indicated for each scale. A difference between *Importance* and *Preparation* is evident.

Item means and standard deviations are presented for each item of the questionnaire for both scales (see Table 3). It can be seen that the mean for *Preparation* was invariably lower than the mean for *Importance*.

Of the 29 items considered most important (M_g 4.5 or above on the *Importance* scale), the majority were from the Instructional Process category. On the other hand, of the 22 items for which the teachers felt they had received better preparation (M_g 2.9 or above on the *Preparation* scale), only a few were from this category.

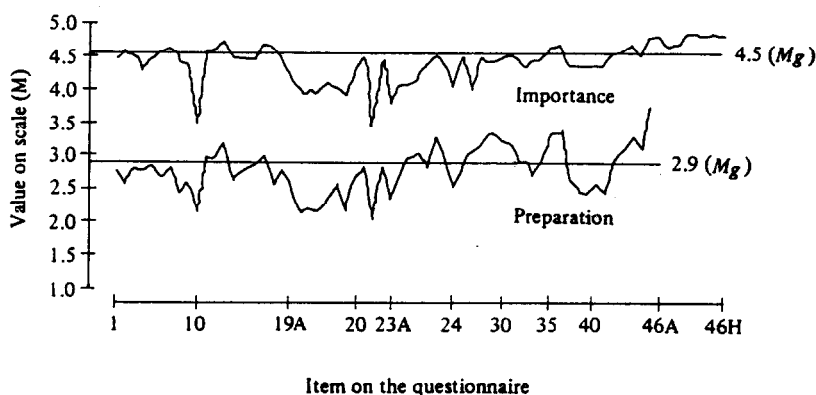


Fig. 1. Mean of each item on the questionnaire and the grand mean (M_g) of each scale.

Table 3

Summary Statistics for Each Item on Both Scales of the Questionnaire

Category	Item No.	n	Importance		Preparation		r*
			M	SD	M	SD	
Instructional process	1	186	4.59	0.75	2.83	1.45	13.80
	2	186	4.66	0.68	2.66	1.49	16.15
	3	177	4.55	0.84	2.85	1.42	14.42
	4	172	4.33	0.90	2.81	1.38	13.28
	5	185	4.55	0.77	2.91	1.36	14.89
	6	184	4.65	0.71	2.74	1.41	16.33
	7	189	4.70	0.63	2.89	1.39	16.47
	8	181	4.50	0.82	2.52	1.42	16.50
	9	184	4.48	0.80	2.63	1.45	15.32
	10	164	3.59	1.21	2.21	1.33	11.17
	11	191	4.65	0.72	3.05	1.32	14.74
	12	192	4.66	0.68	2.99	1.36	15.07
	13	190	4.78	0.50	3.25	1.30	15.44
	14	189	4.49	0.79	2.67	1.45	16.28
	15	185	4.52	0.72	2.82	1.35	16.04
	16	185	4.50	0.77	2.93	1.38	14.85
	17	186	4.75	0.54	3.07	1.36	15.37
	18	182	4.65	0.66	2.60	1.41	17.71
	19	93	4.44	0.88	2.84	1.40	10.86
	20	173	4.35	0.97	2.84	1.35	12.12
	21	180	4.54	0.78	2.88	1.31	15.07
	22	166	3.54	1.16	2.10	1.32	12.68
	23	128	4.55	0.79	2.96	1.43	12.33
	24	170	4.11	1.09	2.56	1.42	12.14
	25	183	4.58	0.68	2.85	1.42	15.35
Foundations	26	182	4.09	1.01	3.12	1.26	8.89
	27	184	4.59	0.78	3.24	1.23	13.53
	28	183	4.44	0.86	3.40	1.18	10.93
	29	185	4.52	0.85	3.29	1.25	12.49
	30	188	4.61	0.71	3.24	1.21	14.03
	31	178	4.54	0.72	2.97	1.29	14.17
	32	189	4.42	0.81	2.99	1.31	13.47
	33	182	4.49	0.83	2.75	1.36	15.46
	34	185	4.57	0.78	2.96	1.33	14.88
	35	190	4.68	0.64	3.39	1.23	13.77
	36	186	4.73	0.57	3.40	1.37	12.55
Other professional attitudes and activities	37	182	4.43	0.72	2.70	1.34	15.54
	38	178	4.42	0.89	2.52	1.33	16.64
	39	182	4.42	0.80	2.52	1.32	16.23
	40	180	4.39	0.92	2.62	1.37	14.40
	41	180	4.44	0.88	2.49	1.41	17.34
	42	182	4.60	0.70	2.95	1.32	15.44
	43	186	4.66	0.62	3.16	1.33	13.99
	44	185	4.76	0.53	3.31	1.25	14.81
	45	183	4.57	0.80	3.14	1.34	13.82
	46	160	4.88	0.36	3.79	1.25	11.13

*Note: $p < 0.05$ for all comparisons.

T-tests indicated significant differences between the way respondents rated each item in terms of the two scales. In all cases, the results showed that the importance of all tasks was

rated significantly higher than the preparation received at the university to perform them ($p < 0.001$).

Content Analysis of Open-ended Questions

All responses to each of the five questions were cumulated. A content analysis was done by key word and key phrase for each question. The results were used to create broad categories. Two judges sorted the responses into these final categories.

The results were summarized as follows.

1. The most outstanding aspect of studies in the Faculty of Education were the courses in academic specialization, and the inspiration generated to carry on the important work of the teacher.

2. The weakest aspects of preparation were the instructional process itself and the ineffectiveness of the Faculty members as teachers.

3. Lacunae in the program were identified as insufficient consideration given to the practical aspects of instruction (the instructional process once more), and on the inadequacy of the practice teaching component of the curriculum.

4. Factors which impeded a better preparation centered, first and foremost, on the competence and attitudes of the professors. This included poor class preparation and serious absenteeism.

5. This question requested that the respondents indicate any omissions in the questionnaire. Instead, it elicited further comments about their preparation. The need for professional development was expressed strongly. Once more the competence of the professors was questioned, the negative role politics played in university life was emphasized and the need for more practice teaching and practical applications of education theories was expressed.

Discussion

In essence, the secondary school teachers validated the curriculum. From their perspective as practicing professionals they rated all items in the curriculum as important. In addition, in the open-ended questions, none suggested there were gaps in the program content. There seemed to be no obvious needs demanded by their work that could not be met by the content of the curriculum. Finally, at no point was the structure of the curriculum questioned. A num-

ber of the open-ended questions could have elicited that information but no one raised the issue. To the teachers, practicing their profession in specific social, political, and economic conditions, and in a particular philosophical context, content and structure did not seem to raise problems of substance.

If what was taught did not appear to be a real concern, how it was taught did. The teachers felt very strongly that they had been poorly prepared by the Faculty of Education. As indicated by the results, the most serious problem in the preparation of teachers was in the Instructional Process category. This did not imply that all was well necessarily in the other two areas, but that the most serious problem was perceived to be in the preparation of those specific teaching skills essential to the practitioners, such as instructional methods and techniques of evaluation.

The responses to the open-ended questions confirmed this conclusion and pointed to possible causes of the problem. The teachers repeatedly mentioned their lack of preparation in the actual teaching process, the lack of practical application of theory, as well as the weakness of the practice teaching component of their program. The professors of the Faculty of Education were made to shoulder a good part of the blame for the shortcomings of the training process. The respondents stated that the professors lacked the competence to teach and that furthermore, they displayed poor attitudes toward their students. In addition, the lack of resources (libraries, laboratories, classroom space, etc.) were blamed for the poor quality of teacher preparation. While the lack of these resources is a reflection of the poverty of the country and of the inadequate funds allocated to the universities, it could be also due to the poor management of the resources and funds available. Another factor identified as contributing to the poor preparation of teachers was politics. Its effects were reflected in two ways. First, the respondents stated that professorial appointments are based on political considerations. This could account, in part, for the lack of competence of the professors as mentioned by the secondary school teachers. Second, it was stated that politics takes up too much of university time in general, and more specifically, is the cause of strikes, other work stoppages and disturbances.

The results of this survey, which was described earlier as a component of program evaluation, provided the Faculty of Education with hard data as to how their preparation of teachers was seen by their consumers. This data, and particularly its salient points, could be used as guidelines for planning curriculum reforms.

The data suggested that changes in curriculum content and structure were not priority areas. The priorities in order of importance seemed to be: to emphasize the instructional process portion of the curriculum; to improve teaching in the Faculty; to consider a more effective allocation of resources; to minimize the undue influence of politics on university life.

The curriculum and the existing courses could be kept in place, but the content should be taught more effectively with emphasis placed on the practical aspects of the teaching process. This is far more difficult a task than considering structural and content changes! It was recommended that a Faculty committee re-examine the curriculum, the content of existing courses, to identify clearly those elements connected to the instructional process, and take the necessary steps to see that these essential aspects of the curriculum be emphasized in such a way as to leave the students with practical teaching skills. The committee created to achieve these objectives has encountered serious management problems. Should the committee or other similar body be able to deal effectively with these issues, the effectiveness of the Faculty to prepare teachers could be increased substantially.

The issue of the competence and attitude of the professors was more difficult to address. It was suggested that a well-structured professional development program be implemented to improve the knowledge base of the professors and to bring them up to date with more current teaching techniques. This is a long-term project currently underway. It was also suggested that some method for evaluating teaching within the Faculty be instituted. This, given the context, traditions, and politics of the university, was unacceptable to the Faculty.

While it was possible to draw to the attention of Faculty the lack and poor management of existing resources as well as the negative effects of politics on their ability to prepare teachers

adequately, recommendations concerning these matters were avoided due to the differing cultural contexts of the authors and the clients.

The survey produced interesting and useful data about the curriculum of the Faculty of Education and the manner in which it was implemented. It not only showed a serious gap between the actual needs of practitioners and their preparation by the university to perform these tasks but identified also areas of weakness.

The method used to gather information and the task of preparing the survey instrument were made easier due to the characteristics of the education faculty being studied. The curriculum was highly structured and rigid and its professional component comprised only 14 courses, all offered in the same sequence. Course outlines were available for all courses and each contained detailed objectives and course content. This information, coupled with the input from instructors who taught those courses, made it easier to develop the list of competencies representing curriculum goals. It would be more difficult, but not impossible, to prepare such a survey instrument with more complex and sophisticated curricula or in the absence of concrete objectives and content for all courses.

Of course, the survey instrument and the process used had limitations. Only the perceptions of consumers of the program were measured, many of whom may have been away from school and the training program for a number of years. The ability of respondents to recall accurately specific elements of their training might be questioned. Also, the survey was adapted to the specific context of one Ecuadorian University. This limits the generalizability of results to other contexts.

However, the results in general do raise questions about the relationship of preparation of professionals by universities and the real needs of practitioners. Earlier studies by Weston (1982) and Boulianne (1983) revealed gaps in the relationship between preparation and practice for other groups of practitioners. Might this imply that professional schools in general have a problem in training practitioners? More specifically, would studies of other teacher training programs reveal weaknesses similar to those reported in the present study? Is the instructional process component, that area which deals with

specific teaching skills, generally a problem area in teacher training programs?

In the final analysis, the survey provided useful information about the curriculum and its application. The data gave the Faculty insights into its operations and placed it in a better position to consider action to improve its performance. Whether this was the information the Faculty was seeking is uncertain since the data indicated that the crux of the problem was the Faculty's application and execution of the curriculum, and not the curriculum structure itself. In essence, the survey results suggest that the Faculty concentrate on reforming itself and not the curriculum in order to improve its performance in training secondary school teachers.

Today there is a great deal of activity on curriculum reform, particularly to make substantial structural and content changes. Those involved in this process might find it useful to gather data from the consumers first to investigate how much the perceived problems may be due to curriculum content and structure and how much to the poor application and execution of existing curricula. Problems which are the result of poor application and execution require very different solutions from those caused by content and structure. Applying a solution pertinent to a problem is the only way to improve effectiveness of performance.

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